ELG 3125 - Signal & Systems

Lecture 1 - Part 2

Periodic Signals

$x(t) = x(t+T)$, for all t.
$x(t) = x(t+mT)$, $m \in \mathbb{Z}$
Periods: $2T, 3T, \ldots$
} Continuous

Fundamental period T_0 of $x(t)$: smallest +ve value of T.

$x[n] = x[n+N]$
$x[n] = x[n+mN]$, $m \in \mathbb{Z}$
Periods: $2N, 3N, \ldots$

Fundamental period N_0 of $x[n]$: smallest +ve value of N

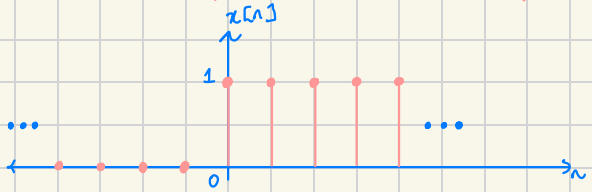
Even and Odd Signals

Even signal: $x(t) = x(-t)$ (symmetric around vertical axis)
Odd signal: $x(t) = -x(-t)$ (symmetric about the origin)

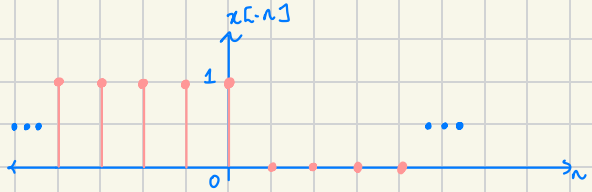
Any signal can be expressed as the sum of its even and odd part.

Continuous	Discrete
$x(t) = Ev\{x(t)\} + Od\{x(t)\}$	$x[n] = Ev\{x[n]\} + Od\{x[n]\}$
$Ev\{x(t)\} = \frac{1}{2}[x(t)+x(-t)]$	$Ev\{x[n]\} = \frac{1}{2}(x[n]+x[-n])$
$Od\{x(t)\} = \frac{1}{2}[x(t)-x(-t)]$	$Od\{x[n]\} = \frac{1}{2}(x[n]-x[-n])$

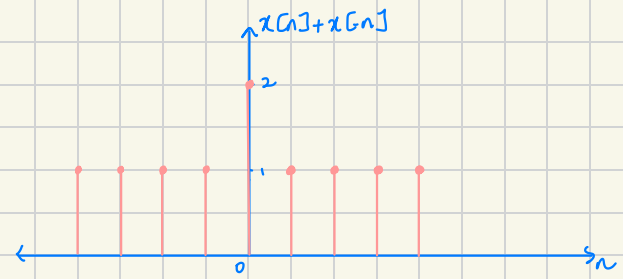
Example: Given $x[n]$, find the even and odd parts of the signal.



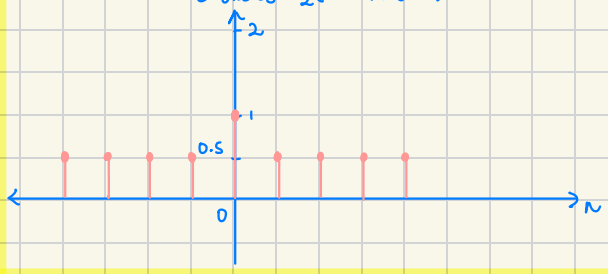
$$x[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

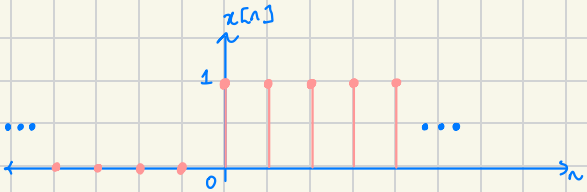


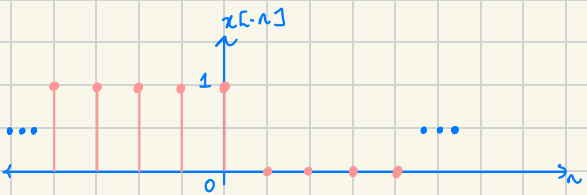
$$Ev\{x[n]\} = \frac{1}{2}(x[n]+x[-n])$$

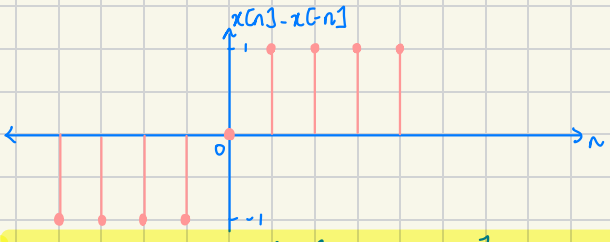


$Ev\{x[n]\} = \frac{1}{2}(x[n]+x[-n])$

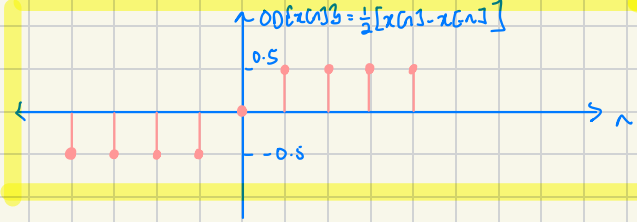






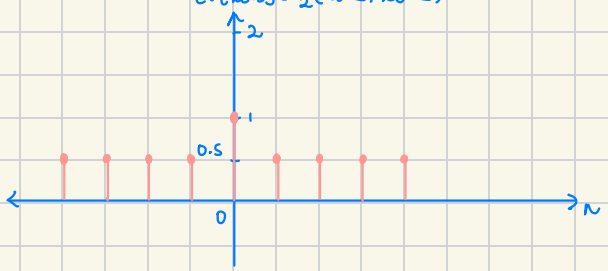


$OD\{x[n]\} = \frac{1}{2}[x[n]-x[-n]]$



Let's Graphically check the fact: $x[n] = Ev\{x[n]\} + Od\{x[n]\}$

$Ev\{x[n]\} = \frac{1}{2}(x[n] + x[-n])$



$Od\{x[n]\} = \frac{1}{2}[x[n] - x[-n]]$

